

AI-Based College Complaint Prioritisation System

Project Documentation & GitHub Reference

Live Project: <https://campus-voice-smart.base44.app>

Project Type	AI-Assisted Web Application
Domain	Education / Complaint Management
Primary Users	Students and College Administrators
Development Platform	Base44 AI
Deployment	Base44 Web Hosting

This document describes the purpose, architecture, workflow, features, AI-based prioritisation concept, testing approach, limitations, and future enhancements of the project.

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Repository note: Add this PDF to your GitHub repository as the project's formal documentation/report.

1. About the Project

The AI-Based College Complaint Prioritisation System is a smart complaint-management platform designed to improve how student complaints are submitted, organized, prioritized, and handled within an educational institution. In traditional college environments, complaints may be submitted through paper forms, emails, suggestion boxes, or informal communication channels. This can make it difficult for administrators to identify urgent issues and ensure important complaints are handled on time.

The proposed platform provides a centralized digital channel where students can submit complaints and administrators can manage them efficiently. AI-based logic can assist in analyzing complaint information and assigning an appropriate priority level.

The application was developed using Base44 AI, an AI-powered application development platform that supports natural-language application development, user interfaces, backend logic, data management, authentication, and deployment.

2. Problem Statement

Colleges receive many different types of student complaints, including:

- Academic issues
- Infrastructure problems
- Hostel-related complaints

- Transportation issues
- Examination-related concerns
- Faculty-related issues
- Laboratory problems
- Internet and technical problems
- Campus facilities and maintenance issues

When these complaints are managed manually, several problems may occur:

- Important complaints may be overlooked.
- Urgent issues may not receive immediate attention.
- Complaints can become difficult to track.
- Students may not know the status of their complaints.
- Administrators may spend significant time manually sorting complaints.
- There may be no systematic method for determining complaint priority.

Therefore, a centralized and intelligent complaint-prioritisation system can improve the efficiency, transparency, and responsiveness of complaint management.

3. Objectives

- Provide students with an easy platform for submitting complaints.
- Digitally organize college complaints.
- Categorize complaints according to their type.
- Identify complaints requiring urgent attention.
- Assist administrators in prioritizing complaints.
- Provide better complaint tracking and management.
- Reduce manual effort in complaint handling.
- Improve transparency between students and college administration.
- Demonstrate the use of AI in solving a real-world educational problem.

4. Proposed Solution

The proposed system provides a centralized complaint-management platform. The general process is:

Student	→	Submit Complaint	→	Complaint Stored
Complaint Stored	→	AI Analysis	→	Category + Priority
Category + Priority	→	Administrator Dashboard	→	Resolution
Resolution	→	Status Update	→	Student Tracking

The system helps administrators focus on complaints according to their importance instead of treating every complaint with the same priority.

5. Key Features

Student Features

- Submit college complaints online.
- Provide complaint details.
- Select or provide relevant complaint information.
- View submitted complaints.

- Track complaint status.
- Access the system through a web browser.

Administrator Features

- View submitted complaints.
- Monitor complaints through a centralized dashboard.
- Identify high-priority complaints.
- Manage complaint categories.
- Update complaint status.
- Track the progress of complaint resolution.

AI-Based Features

- Analyze complaint information.
- Assist with complaint categorization.
- Determine an appropriate priority level.
- Help administrators identify urgent complaints.
- Reduce the amount of manual sorting required.

6. AI-Based Prioritisation

The core concept of the project is to use AI to assist in determining the priority of a complaint. A complaint can be evaluated using factors such as:

- Urgency
- Severity
- Number of students affected
- Type of issue
- Potential impact on academic activities
- Safety or infrastructure impact

Conceptual prioritisation pipeline:

Stage	Description
1	Analyze complaint details
2	Identify complaint category
3	Evaluate urgency and impact
4	Assign priority
5	Classify as High / Medium / Low

Complaint Example	Possible Priority
Broken classroom fan	Medium
Campus Wi-Fi unavailable	Medium
Examination portal not working	High
Major water supply failure	High
Minor classroom maintenance request	Low

Note: The exact AI prioritisation behavior depends on the implementation and configuration of the Base44 application. AI output should be treated as decision support, with administrators retaining final responsibility for complaint handling.

7. Technology Used

Layer	Technology / Approach
Frontend	Web-based responsive user interface, forms, and dashboard components
Backend	Base44 application backend, database/data management, and application logic
Authentication	Authentication and access management where configured
AI	Base44 AI-assisted application development and complaint analysis/prioritisation
Hosting	Base44 web hosting and live application deployment

8. System Modules

1. User Authentication Module — Provides controlled access to the application for users.
2. Complaint Submission Module — Allows students to submit complaints by providing relevant details.
3. Complaint Management Module — Stores and organizes submitted complaints.
4. AI Prioritisation Module — Analyzes complaint information and assists in determining priority.
5. Admin Dashboard — Allows administrators to monitor and manage complaints.
6. Complaint Status Module — Allows complaints to be tracked through different stages of resolution.

9. User Roles

Role	Capabilities
Student	Access system; submit complaints; view submitted complaints; track complaint status
Administrator	View complaints; analyze priorities; manage complaints; update status; monitor unresolved issues

10. System Workflow

The end-to-end workflow can be represented as follows:

Student	↓
Submit a Complaint	↓
Store Complaint Data	↓
AI Analysis — Category + Priority	↓
Administrator Dashboard	↓
Resolve Complaint	↓
Update Status	↓
Student Tracks Status	

11. Advantages

- Centralized complaint management.
- Faster identification of urgent complaints.
- Reduces manual complaint sorting.
- Improves transparency.
- Easy complaint tracking.
- Better organization of complaints.
- Helps administrators manage workload.
- Accessible through the web.
- Demonstrates a practical application of AI in education.
- Can be expanded for use across different departments.

12. Applications

The system can be used in engineering colleges, universities, degree colleges, schools, hostels, and campus administration departments.

It can be adapted for complaints related to:

- Infrastructure
- Academics
- Transportation
- Hostels
- Laboratories
- IT services
- Campus security
- General student services

13. Limitations

- AI-generated priorities may require human verification.
- Priority accuracy depends on the quality of complaint information.
- The system requires internet access.
- Advanced integrations may require additional configuration.
- The application may need further testing with large volumes of real-world complaints.

AI prioritisation should therefore be treated as decision support, while administrators retain final responsibility for complaint handling.

14. Future Enhancements

- Email notifications
- Mobile application support
- Real-time complaint notifications
- Advanced analytics dashboard
- Complaint trend analysis
- Improved AI classification models
- Multilingual complaint submission
- File and image attachments

- Automated chatbot support
- Department-based complaint routing
- SLA/deadline tracking
- Advanced role-based access control
- Complaint reports and PDF export
- Monthly and yearly performance reports
- Automatic escalation of unresolved high-priority complaints

15. How to Use

Step 1 - Open the Application — Visit <https://campus-voice-smart.base44.app>

Step 2 - Access the Application — Use the authentication/access mechanism provided by the application.

Step 3 - Submit a Complaint — Enter the required complaint information and submit the complaint.

Step 4 - AI-Based Prioritisation — The application processes the complaint and assists in identifying its category and priority.

Step 5 - Administrator Review — The administrator reviews the complaint and takes appropriate action.

Step 6 - Update Status — The complaint status is updated as the issue progresses toward resolution.

16. Project Structure

If the Base44 project is synchronized or exported to GitHub, the exact generated structure may vary. A typical repository can be organized as:

```
ai-based-college-complaint-prioritisation/
├── README.md
├── src/
│   ├── components/
│   ├── pages/
│   ├── services/
│   ├── public/
│   ├── assets/
│   ├── database/
│   ├── .gitignore
│   └── package.json
```

The actual folder structure should be updated after exporting/synchronizing the Base44 project to GitHub.

17. Development Methodology

Phase 1 - Requirement Analysis — The problem of inefficient college complaint management was identified.

Phase 2 - System Design — Users, complaint workflow, prioritisation process, and dashboard requirements were defined.

Phase 3 - AI-Assisted Development — Base44 AI was used to generate and refine the application using natural-language requirements.

Phase 4 - Interface Development — The application interface, complaint forms, dashboards, and user flows were developed and refined.

Phase 5 - Testing — The application was tested using different complaint scenarios and workflow checks.

Phase 6 - Deployment — The application was published as a web application using Base44 hosting.

18. Testing

Test Case	Input / Action	Expected Result
TC01	Student submits complaint	Complaint is recorded
TC02	High-impact complaint submitted	Complaint receives high priority/attention
TC03	Normal maintenance complaint	Complaint receives lower priority
TC04	Administrator opens dashboard	Complaints are displayed
TC05	Administrator updates status	Updated status is reflected
TC06	Student checks complaint	Complaint status can be viewed

19. Expected Outcome

- Simplifies complaint submission.
- Organizes complaints efficiently.
- Helps identify urgent complaints.
- Reduces administrative workload.
- Improves complaint tracking.
- Provides a better communication channel between students and administration.

20. Security and Privacy

The application should be configured to ensure that:

- Users only access information appropriate to their role.
- Complaint data is protected.
- Administrative functions are restricted to authorized users.
- Sensitive student information is not unnecessarily exposed.

Base44 provides authentication, access controls, and security-related capabilities that can be configured as part of an application.

21. Screenshots

Add screenshots of the deployed application to the GitHub repository to visually document the system. Recommended screenshots include:

- Home / landing page
- Student complaint submission page
- Student complaint tracking page
- Administrator dashboard
- Complaint details and priority view
- Complaint status update screen

Example Markdown for README.md:

```
## Screenshots
### Home Page
![Home Page](screenshots/home.png)
```

22. Conclusion

The AI-Based College Complaint Prioritisation System demonstrates how AI-assisted application development can be applied to a practical educational problem. By providing a centralized platform for complaint submission, categorization, prioritisation, tracking, and administration, the system can help colleges respond to important issues more efficiently.

The project also demonstrates the value of combining a web-based workflow with AI-assisted decision support. Although the current implementation is a prototype/mini-project and requires human verification of AI-generated priorities, it provides a strong foundation for future enhancements such as analytics, notifications, multilingual support, department routing, SLA tracking, and automated escalation.

23. Acknowledgements

This project was developed with the assistance of Base44 AI, an AI-powered application development platform used for application generation, refinement, backend/data capabilities, and deployment. The project documentation is intended to accompany the live application and GitHub repository.

Live Application: <https://campus-voice-smart.base44.app>